

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	C. Shahanaj	Reg. No.:	192372292
Section / Batch:	CSE-A1/2023	Date:	25/06/2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions

Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1
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SECTION A – CONCEPT MAPPING

Code of Conduct

I Shaharaj- 190372292 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Shaharaj

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established

Organizational	20%	Concepts are well structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%				
Linkages	25%				
Organization	20%				
Integration of Literature	20%				

Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Recurrence Relation

(e.g., $T(n) = 2T(n/2) + n$)

Formulate initial Guess

- Derive candidate bound using Recursion Trees, Iterative Expansion, or Asymptotic Intuition.
- Target: Bound of the form $O(g(n))$.

Induction Hypothesis

- Assume $T(k) \leq c * g(k)$ holds for all $k < n$.
- RULE: use explicit constants ($c > 0$).
- RULE: Do NOT use Big-O notation in algebra

Inductive step

- Substitute hypothesis into original recurrence:
Replace subproblems with $c * g(n/2)$.
- Algebraically simplify expression toward $c * g(n)$

Does algebra reduce cleanly
to $\leq c * g(n)$?

No

Yes

Refine Guess

- Prbl: Residual positive terms prevent matching.
- Fix: subtract a lower-order term from guess
- Re-run Inductive step with updated guess

Base case verification

- Test boundary: $T(n_0) \leq c * g(n_0)$
 - $n_0 = 1, 2, \text{ or } 3$
- Adjust n_0 if boundary values yield illegal terms

Mathematical correctness proof

- Grounded base case + valid inductive step proves $T(n) \leq c * g(n)$ for ALL $n \geq n_0$ up to infinity
- Final Result: $T(n) = O(g(n))$ formally proven.

Recurrence Relation: $T(n)$



Formulate Guess: $g(n)$



Induction Hypothesis

(Assume $T(k) \leq C \cdot g(k)$ for $k < n$)

Rule: Use constants $C > 0$, No Big-O in algebra



Inductive step

(substitute hypothesis
& simplify to $\leq C \cdot g(n)$)

Base case Verification

prove $T(n_0) \leq C \cdot g(n_0)$ to
anchor the proof



(If stuck: refine guess with -d)



Solve for constants (C & n_0)



Mathematical correctness: $T(n) = O(g(n))$

